

Multiobjective Optimization

A-posteriori Methods: Concepts and Solution Generation

Bruno de Athayde Prata

Preference Articulation Categories

- **A priori methods:** Preferences specified before optimization
- **Interactive methods:** DM involved throughout the process
- **A posteriori methods:** First generate the entire PF, then select

Key Characteristics of A-posteriori Methods:

- Generate the entire (or representative set of) PF first
- Decision maker evaluates solutions in post-processing phase
- Final choice converges to an acceptable compromise
- An MCDA approach guides DM toward preferred points

Source: Ehrgott et al. (2026), Section 2

Decision and Objective Spaces

Decision Space (X):

- Feasible set in decision variables
- Contains all possible values of decision variables
- Decision vector: $x \in X$

Objective Space (Z):

- Image of X in objective function space
- All possible objective function values
- Objective vector: $z(x) = (z_1(x), z_2(x), \dots, z_p(x)) \in Z$

Mapping:

- If $x \in X$, then $z(x) \in Z$
- All objectives assumed to be minimized

Source: Ehrgott et al. (2026), Section 2

Dominance:

- $z(x) \in Z$ is **dominated** if $\exists z(x') \in Z$ such that:
 - $z(x') \leq z(x)$ (at least as good in all objectives)
 - $z_j(x') < z_j(x)$ for at least one j

Non-dominance:

- If no such $z(x')$ exists, $z(x)$ is **non-dominated**
- x is **efficient (Pareto-optimal)**

Key Concepts:

- X_E : Efficient set in decision space
- Z_N : Non-dominated set in objective space
- **Pareto Set (PS)**: Set of all efficient decision vectors
- **Pareto Front (PF)**: Image of PS in objective space

Source: Ehrgott et al. (2026), Section 2

Types of Efficient Solutions

Supported Efficient Solutions:

- There exists a linear scalarization that makes x optimal
- Lie on the convex hull of the PF

Unsupported Efficient Solutions:

- No linear scalarization exists
- Cannot be found by weighted sum method
- Common when feasible set is non-convex or discrete

Why This Matters:

- Weighted sum only finds supported solutions
- Need specialized methods for unsupported solutions
- Important for MOCO and MOIP problems

Source: Ehrgott et al. (2026), Section 2

Key Points for Understanding Trade-offs

Ideal Point (z^*):

- Best possible value for each objective individually
- $z_j^* = \text{Minimize } z_{x \in X} z_j(x), j = 1, \dots, p$
- Usually infeasible
- Serves as a reference for how good an objective can be

Nadir Point (z^N):

- Worst objective values among all Pareto-optimal solutions
- $z_j^N = \text{Maximize } z_{x \in X} z_j(x), j = 1, \dots, p$
- Hard to find for more than three objectives

Reference Point (z^R):

- User-defined aspiration levels
- Used to guide algorithms toward preferred solutions
- Need not be feasible

Source: Ehrgott et al. (2026), Section 2

Problem Classification

By Objective/Constraint Nature:

- **Linear:** All objectives and constraints linear
- **Non-linear:** At least one non-linear function
- **Convex:** Objective functions and feasible region convex
- **Non-convex:** Computationally more challenging

By Decision Variable Nature:

- **Continuous:** Variables can take any value
- **Discrete:** At least one variable restricted to integers
- Typically provably computationally hard

By Problem Evolution:

- **Static:** Parameters do not change
- **Dynamic:** Parameters evolve over time

Source: Ehrgott et al. (2026), Section 2

Two Main Categories

1. Mathematical Programming-Based Approaches:

- Reformulate MOO into single-objective problems
- Use established optimization techniques
- Guarantee certain desirable characteristics
- Find one solution per run for given parameters

2. Population-Based Approaches:

- Mimic natural processes
- Evolve population of multiple solutions simultaneously
- Heuristic in nature
- No optimality guarantees
- Include evolutionary computation and swarm intelligence

Source: Ehrgott et al. (2026), Section 1 & 4

Linear Scalarization

Formulation:

$$\text{Minimize } \sum_{j=1}^p w_j z_j(x)$$

$$\text{s.t. } x \in X$$

$$w_j \geq 0, \quad \sum_{j=1}^p w_j = 1$$

Properties:

• Pros:

- Simple and intuitive
- For convex PF: Can find all supported efficient solutions

• Cons:

- Cannot find unsupported efficient solutions
- Uniform weights $\neq$ uniform PF distribution
- Fails for non-convex PF

Source: Ehrgott et al. (2026), Section 3.1

Formulation and Properties

Basic Formulation:

$$\begin{aligned} &\text{Minimize } z_1(x) \\ &\text{s.t. } z_i(x) \leq \varepsilon_i, \quad i = 2, \dots, p \\ &x \in X \end{aligned}$$

Guaranteeing Efficiency:

$$\text{Minimize } z_1(x) + \rho \sum_{i=2}^p z_i(x)$$

where $\rho > 0$ is sufficiently small

Advantages:

- Can find supported and unsupported solutions
- Works for any PF shape
- Controllable via ε values

Source: Ehrgott et al. (2026), Section 3.1

Reference Point Approach

Key Concept:

- Generalizes Chebyshev distance functions
- Projects any point in objective space onto the efficient frontier
- Can reach any efficient solution with appropriate parameters

Basic Form:

$$\begin{aligned} &\text{Minimize } \max_i \{w_i(z_i(x) - z_i^R)\} + \rho \sum_{i=1}^p w_i z_i(x) \\ &\text{s.t. } x \in X \end{aligned}$$

Properties:

- Handles non-convexity naturally
- Reference point z^R guides the search
- Weights w_i determine search direction

Source: Ehrgott et al. (2026), Section 3.1

Why Evolutionary Computation for MOO?

- ① Use **population of solutions** in each iteration
 - Ideal for finding multiple solutions simultaneously
- ② **Implicit parallel search**
 - Sharing common properties can speed up discovery
- ③ **Flexible and generic**
 - No gradient or auxiliary information needed
 - Only objective and constraint function values required

Main Limitation:

- Approximate PF rather than systematically proving optimality
- Due to randomized operators and finite computational resources

Source: Ehrgott et al. (2026), Section 4.1

Key Components

1. Non-dominated Sorting:

- Sort population into different non-dominated fronts
- First front: all non-dominated solutions
- Second front: dominated only by first front, etc.

2. Crowding Distance:

- Measure of solution density in objective space
- Larger distance = more isolated solution
- Preserves diversity in the population

3. Elitism:

- Merge parent and offspring populations
- Select best individuals for next generation
- Preserves top-performing solutions

Limitation: Does not scale well for more than three objectives

Source: *Ehrgott et al. (2026), Section 4.2.1*

Core Concept:

- Decompose MOO problem into scalar subproblems
- Each subproblem defined by weight vector/aggregation function
- Solve subproblems simultaneously

MOEA/D (Zhang & Li, 2007):

- Pioneering decomposition-based EMOA
- Uses neighborhood structure for information exchange
- Highly efficient, many variants exist

NSGA-III (Deb & Jain, 2014):

- Extends NSGA-II with reference vectors
- Designed for many-objective problems
- Provides balanced distribution for non-convex/discontinuous fronts

Source: Ehrgott et al. (2026), Section 4.2.3

Three Main Groups

Category	Key Feature	Representative
Pareto-Based	Uses Pareto dominance for ranking	NSGA-II, SPEA2
Indicator-Based	Uses quality indicators to guide search	IBEA, SMS-EMOA, HypE
Decomposition-Based	Decomposes into scalar subproblems	MOEA/D, NSGA-III

Key Observations:

- Pareto-based: Most popular, but struggle with many objectives
- Indicator-based: Good convergence, but computationally expensive
- Decomposition-based: Prevailing trend, scalable to many objectives

Source: Ehrgott et al. (2026), Section 4.2

Four Groups of Metrics (Audet et al., 2021):

- ① **Cardinality Indicators:** Number of non-dominated solutions
- ② **Convergence Indicators:** Closeness to true PF
- ③ **Distribution/Spread Indicators:** Distribution of solutions
- ④ **Hybrid Indicators:** Combine multiple properties

Key Hybrid Metrics:

- **Hypervolume (HV):**
 - Volume of objective space covered by PF approximation
 - Favorable mathematical properties
- **Inverted Generational Distance (IGD):**
 - Average distance from true PF to nearest generated solution
 - Combines convergence and diversity

Source: Ehrgott et al. (2026), Section 4.3

Benchmark Problems in EMO

ZDT Suite:

- Two objectives
- Varying PF shapes (convex, concave, disconnected)
- Tests convergence and diversity

DTLZ Suite:

- Scalable to many objectives
- Tests scalability of algorithms
- Linear, convex, concave, disconnected PFs

WFG Suite:

- Complex landscapes
- Multi-modality
- Non-convex PFs

Recent Suites:

- **MaF**: Many-objective problems
- **MW**: Multi-modal problems
- **DASCOMP**: Constrained problems

Overview of Application Areas

Engineering Design:

- Structural optimization (trusses, composite plates)
- Turbine and fan blades, pumps
- Heat exchangers, suspension systems

Finance & Economics:

- Portfolio optimization (return, risk, ESG)
- Financial time series, stock ranking
- Trading decision-support tools

Energy:

- Wind farm layouts, solar plants
- Energy distribution networks
- Electricity market optimization

Manufacturing:

- Job-shop and flow-shop scheduling
- Resource allocation
- Production planning

Key Differences

Feature	Mathematical Programming	Population-Based
Optimality Guarantee	Yes	No (approximate)
Single Run	One solution	Multiple solutions
Unsupported Solutions	Can find	Can find
Non-convex PF	Depends	
Scalability	Limited	Better
Gradient Information	Often needed	Not needed
Problem-Specific	Tailored	Generic

Hybrid Approaches:

- Combine strengths of both methods
- EMO for global search + Mathematical for local refinement
- Interactive frameworks with EMO and MCDM

Source: Ehrgott et al. (2026), Section 6

Preference Information Types

Direct Preference Information:

- Reference points (aspiration levels)
- Reference directions
- Weights (importance coefficients or trade-offs)
- Comparison thresholds

Indirect Preference Information:

- Pairwise comparisons of solutions
- Ranking of a small set of solutions
- Classification into preference-ordered classes
- Scoring of solutions

Trend: Shift toward indirect preferences

- More natural for most DMs
- Reduced cognitive effort
- Less knowledge of problem required

Source: Ehrgott et al. (2026), Section 5.1

Preference Acquisition Modes

1. A Priori:

- Preferences specified before optimization
- Search focused on satisfactory solutions
- May be looped (semi-interactive)

2. Semi-Interactive (IAR):

- Interaction after a complete run
- Analyze solutions, revise preferences
- Rerun algorithm to find more relevant solutions

3. Progressive Interactive (IDR):

- Interaction during the run
- Single run interleaves search and decision-making
- Bias population toward DM's ROI
- Most common for indirect preferences

4. A Posteriori:

- DM selects from generated PF approximation
- Overview of various solutions
- Faster interaction at cost of computational effort

Three Families of Models

1. Scoring Functions:

- **Weighted Metrics:** Distance to reference point
 - Chebyshev functions, Euclidean distance
 - Achievements scalarizing functions
- **Value Functions:** Comprehensive desirability measure
 - Additive linear, piecewise linear, general monotonic
 - Choquet integral, Gaussian, polynomial

2. Binary Relations:

- Outranking relations (ELECTRE, PROMETHEE)
- Dominance enrichment (g-dominance, r-dominance)
- Robust preference relations

3. Decision Rules:

- "If ... then ..." statements
- Fuzzy inference systems
- Decision trees
- Explainable models with natural syntax

Source: Ehrgott et al. (2026), Section 5.3

1. Many-Objective Optimization:

- More than 3 objectives
- Scalability challenges
- Dimensionality reduction
- Visualization difficulties

2. Machine Learning Integration:

- Surrogate models for expensive evaluations
- Neural networks for preference learning
- Generative AI for solution generation
- Automated discovery of optimization strategies

Source: Ehrgott et al. (2026), Section 6

3. Uncertainty Handling:

- Robust optimization
- Stochastic programming
- Dynamic problems
- Uncertainty quantification

4. Personalization:

- Tailored solutions to specific user needs
- Behavioral insights
- Cognitive load reduction

Source: Ehrgott et al. (2026), Section 6

Preference Acquisition:

- Effective elicitation techniques
- Cognitive effort reduction
- Behavioral biases understanding

Performance Assessment:

- Benchmarking guidelines
- Evaluation of interactive methods
- Many-objective metrics

Scalability:

- High-dimensional decision spaces
- Large-scale instances
- Many objectives

Practical Applications:

- Bridge theory-practice gap
- Real-world testing
- Decision support systems

Technology-Driven Opportunities

Quantum Computing:

- Potential to address combinatorial and high-dimensional challenges
- Hybrid quantum-MOO approaches
- Faster convergence and exploration of decision spaces

Artificial Intelligence:

- Neural networks and deep learning
- Generative AI for problem modeling
- Automated preference elicitation

Visualization:

- Enhanced tools for evaluations
- Interactive decision support
- Preference elicitation improvements

GenAI in MOO:

- Automating identification of relevant objectives and constraints
- Training on individual's past choices
- Filtering solutions based on preferences
- Collective judgment analysis

Major Problems Facing Humanity:

- Climate change
- Poverty and famine
- Regional conflicts
- Threats to democracies

Why MOO Matters:

- These are multifaceted, complex problems
- Studying them from multiple perspectives is crucial
- Structured approaches can bring fresh ideas
- Hope for a better world

"MCDM researchers have important tools to meet these challenges." — Ehrgott et al. (2026)

Source: Ehrgott et al. (2026), Section 6.2

Key Takeaways

1. A-posteriori Methods:

- Generate solutions first, choose later
- Allow DMs to see full range of trade-offs

2. Two Main Approaches:

- Mathematical programming: Exact, single solution per run
- Evolutionary computation: Approximate, multiple solutions per run

3. Algorithm Classes:

- Pareto-based, indicator-based, decomposition-based

4. Preference Incorporation:

- Direct and indirect preferences
- Various interaction patterns
- Three preference model families

5. Future Directions:

- Many-objective optimization
- Machine learning integration
- Quantum computing

Core Reference:

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